Earthscan Oceans

MARINE POLICY

An Introduction to Governance and International Law of the Oceans

Mark Zacharias and Jeff Ardron

SECOND EDITION

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This newly updated book shows how the various approaches to ocean governance can be integrated to improve social, economic, and environmental outcomes. It should be essential reading for all those of us who strive to better manage and protect our home-planet ocean.

Dan Laffoley, Marine Vice Chair, International Union for Conservation of Nature (ICUN) World Commission on Protected Areas

Citing examples from around the world, this book provides a timely and useful perspective on the challenges facing effective marine policy, including the complexities of climate change, along with new and emerging issues such as ocean energy and deep-sea mining.

Jon C. Day, James Cook University, Australia and formerly Director with the Great Barrier Reef Marine Park Authority

Marine Policy

This book provides readers with a foundation in policy development and analysis, describing how policy, including legal mechanisms, are applied to the marine environment. It presents a systematic treatment of all aspects of marine policy, including climate change, energy, environmental protection, fisheries, mining and transportation.

The health of marine environments worldwide is steadily declining, and these trends have been widely reported. *Marine Policy* summarizes the importance of the ocean governance nexus, discussing current and anticipated challenges facing marine ecosystems, human activities, and efforts to address these threats. This new, fully revised edition has been updated throughout, including content to reflect the recent advances in ocean management and international law. Chapters on shipping, energy/mining and integrated approaches to ocean management have been significantly reworked, plus completely new chapters on the United Nations Convention on the Law of the Sea, and the impacts of climate change have been added. Pedagogical features for students are included throughout.

Aligned with current course offerings, this book is an ideal introduction for undergraduates and graduate students taking marine affairs, science and policy courses.

Mark Zacharias is the Deputy Minister of the Ministry of Environment and Climate Change Strategy with the province of British Columbia, Canada. He is also Adjunct Professor at the University of British Columbia Fisheries Centre and Associate Adjunct Professor in the Department of Geography at the University of Victoria. He is the co-author of the book *Marine Conservation Ecology* (Earthscan, 2011).

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Marine Policy

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Mark Zacharias and Jeff Ardron

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Preface to the second edition

In his classic text *Unpopular Essays*, Bertrand Russell contended that humanity confronts two different problems: The first is mastering nature to provide for human needs, which is firmly in the domain of science and technology (Russell, 1950). The second is wisely utilizing the fruits of science and technology to improve the human condition. Prudence, however, has not always prevailed. Russell provided examples of technological advances throughout history that led to disastrous outcomes. In particular, he evoked the domestication of the horse (conquest), mechanization (slavery) and modern physics (nuclear weapons) as specific advances that were accompanied by unfortunate consequences. Russell stated that while science and technological achievements and skills are important, 'something more than skill is required, something which may perhaps be called "wisdom". This is something that must be learnt, if it can be learnt, by means of other studies than those required for scientific technique'.

Russell's observations are especially apropos to our relationship with the ocean. Evidence confirms that humans consumed seafood 164,000 years ago, and early pre-Western cultures developed sufficient technical expertise to overharvest species they could access. The advent of industrialization created the opportunity to fish in remote oceans, extract oil and gas from continental shelves, transport goods over great distances and use the ocean as a repository for waste. Yet, even as late as the early 20th Century, many still thought the ocean immutable and immune to human activities.

Paralleling Russell's examples, scientific and technological advancements have outpaced humanity's ability to understand the consequences of our activities on the marine environment. Nor have we been able to envision how to regulate these activities as a whole. Even after evidence emerged on the effects of overfishing and pollution, piecemeal responses took decades. This unfortunate delay is exacerbated by the fact that, even to this day, most of the ocean is owned by no one and thus must be managed as a commons resource where the negligence of a few can come at the expense of many: a tragedy of the commons in general, but specifically a tragedy of international cooperation. In every international forum that we have visited or participated in, we have witnessed a great many states that wish to improve the lot of the ocean, only to be thwarted by an intransigent minority who have held other interests.

This book explores what Russell termed the 'wisdom' necessary to ensure that humanity's scientific and technical achievements – and the consequences of these achievements – lead to what he termed 'contentment'. In the context of the nine years of difficult negotiations that led to the signing of the Law of the Sea Convention (LOSC) in 1982, the word 'contentment' might have been perhaps described then as 'peaceful equity' – an ocean free of conflict, with resources available for all of humanity. More than 30 years later, the peace has held – a testament to the international diplomacy of that time. However, the question of equity is not so easily answered. For some states, the rich fisheries in the 1970s are but a distant memory. Some continue to subsidize their commercial fleet to fish deeper and further away, whilst other states place new hope in an old idea, wagering that deep-seabed mining's time has finally arrived. If Ambassador Arvid Pardo, who urged the United Nations General Assembly in 1967 to develop a fair governance regime for the ocean, were alive today, would he say, 'job well done'?

To be fair, the management of the global ocean is incredibly complex. Even those employed full-time in the field of ocean governance find it difficult to be well-informed in all its aspects. At present, there are over 150 multilateral international agreements that govern the use of the marine environment. In turn, many of these agreements establish dozens of multinational agencies, commissions, organizations and secretariats to oversee their implementation. These administrative bodies then create policies, guidelines and reporting requirements. Increasing the complexity is that many of these agreements nest hierarchically within other agreements or require other agreements to be fully implemented. The blue Byzantine maze wavers and shifts over time, such that its international conventions are frequently amended, or buttressed by key decisions that are not always easy to find. As if this was not enough, most geographic regions of the ocean are governed by multiple, separate regional agreements, which may or may not have the same states as parties.

Unsurprisingly, no single person can have a comprehensive understanding of what might make this international ocean governance more effective. Staring down the double barrels of climate change and ocean acidification, we will need to better listen to one another, trust one another and learn together.

Purpose of this book

This intent of this book is to provide the reader with an understanding of how nation-states currently work together to address socio-economic, environmental and cultural issues with respect to the management of the world ocean. Marine governance is an amalgam of historical practice and custom, economic and trade considerations, domestic law and policy, and regional and international agreements. How oceans are governed is considered here through the lens of 'marine policy'. Related but distinct from maritime law or marine science, marine policy is the stated intent and subsequent actions, usually by governments, to influence ocean-related development, activities and outcomes. In this way, marine policy often includes the use of science and laws to achieve its stated goals.

While this text introduces the legal context and international law that underpins ocean governance, it is not a law textbook nor is it an authoritative guide to any particular sector or convention. Dozens of books have been written on each of the international conventions discussed here and our intent is not to duplicate these scholarly works. Instead, this textbook explores the how various approaches to ocean governance have emerged in various maritime sectors (e.g. fishing or transportation) and the opportunities to learn from, and integrate, these approaches to improve social, economic and environmental outcomes.

This text primarily focuses on the international aspects of marine management. While each nation-state (even landlocked states) has legal standing in international maritime law and may have its own set of marine laws and policies, this book will cast only occasional glances at individual nations and their domestic marine agendas by way of example. There are many excellent 'coastal management' texts, to which the curious reader is directed. However, this book attempts to provoke the reader, who may be a specialist in one particular aspect of marine research, law or policy, towards thinking about the opportunities and benefits of taking a more holistic view.

Why this book now and why a second edition

The current environmental condition of marine environments worldwide is steadily declining, and these trends have been widely chronicled. Traditional, sector-based approaches to ocean governance (such as managing fisheries or regulating transportation) have stumbled when attempting to resolve the bigger social and environmental predicaments facing humanity (see Box 0.1). Climate change, sovereignty claims and regional economic disparity are not addressed by drafting an improved single-sector regulation. Complex, interrelated problems require sophisticated interconnected decision-making, aimed towards common goals.

Dramatic changes in the world of ocean governance have occurred since the publication of the first edition in 2014. For the first time ever, the ocean was recognized as an international priority, and provided with its own development goal and targets, in the form of United Nations Sustainable Development Goal 14. In 2017, the United Nations followed up with its first Ocean Conference since the days of the Law of the Sea negotiations. Meanwhile, climate change has been found to be more rapid and damaging than previously thought, warranting a completely new (nearly) global agreement under the UNFCC. In addition, significant progress has occurred on regional bi- and multilateral agreements to address fisheries management and marine health. The first edition provided only a cursory treatment of marine pollutants, which is rectified in this edition. Finally, the rapid evolution of marine planning approaches continues and is expanded upon in this new edition.

Box 0.1 Governance challenges resulting from the unique characteristics of the marine environment

- Sector-based single-focus solutions are often poor fits for the biophysical complexity of marine environments and interconnections with the economy and human communities.
- Social, economic and ecological interactions at multiple geographical scales are not well understood, but one size (scale) of governance is unlikely to be the solution. Therefore, a cooperative multi-scalar approach is required.
- Management based on 'command and control' regulatory approaches on land is less effective in marine systems because of their unpredictable dynamics and complexity.
- Short-term (human scale) attitudes and timeframes can hobble planning for long-term sustainable solutions compatible with marine ecological timeframes.
- A 'frontier mentality' of a limitless ocean that assumes few ecological, thermodynamic and economic constraints to resource development is centuries-old thinking that does not acknowledge the explosion of human technological power over the past 50 years.
- Marine systems transcend national and regional boundaries, leaving management systems developed for fixed-boundary property rights (ownership) inadequate.

Source: Significantly modified from Glavovic (2008)

Book outline

Until the 20th Century, the regulation of maritime activities was widely perceived as unnecessary and unachievable given the widespread belief that the vast ocean was impervious to human efforts. **Chapter 1** summarizes the importance of the world ocean to humanity and the modern human footprint on the world ocean. The chapter then discusses threats to biodiversity, which can be broadly categorized as a result of overharvesting, pollution, habitat loss, introduced species, ocean acidification and global climate change.

The discrete inner workings of the oceans are largely invisible to us and often counter-intuitive to our terrestrial experience. As such, a basic knowledge of the biological and physical characteristics of marine systems is required to understand how the application of marine law and policies may impact marine environments. An 'Oceans 101' is provided in **Chapter 2** to familiarize the reader with the structure, function and processes that operate in marine environments. It begins with an exploration of the differences between terrestrial and marine environments. The intent of this section is to evince how our 'land referenced' perspective instils a bias that must be overcome if we are to fully fathom how marine environments could be governed.

Chapter 3 introduces the history and operation of international law. Within this chapter is a discussion of the types of ocean ownership regimes, an introduction to the world's different legal systems and a brief explanation of the workings of international law. The chapter then introduces international marine law and the differences between marine law and other legal systems. It concludes with a summary of international marine conventions and non-binding agreements.

Chapter 4 introduces the United Nations Convention on the Law of the Sea (LOSC). The LOSC is one of the most comprehensive international agreements created, with an extensive literature all its own. This chapter provides an overview of the history, content and operation of the LOSC while subsequent chapters will delve further into the specific functions of the Convention.

An introduction to the field of policy encompasses **Chapter 5**. It explores the purpose of public policies, types of policies and the relationship between law and public policy. The unique characteristics of marine law and policy are then discussed with particular attention to areas beyond national jurisdiction (the high seas and deep seabed). The fifth chapter then reviews how policies are formed, the characteristics of policy development, considerations in selecting policies and methods for selecting and analysing policies. Central to this textbook, this chapter is its longest.

The primary international mechanisms that govern the protection and preservation of the oceans – along with the important regional conventions and examples of bi-lateral and multilateral agreements – are presented in **Chapter 6**. The chapter outlines the primary international and regional agreements governing the protection of marine environments.

Addressing climate change and its impacts on the world ocean is the subject of **Chapter 7**. The chapter first reviews the impacts of climate change on the oceans and then discusses the international efforts to mitigate and adapt to a changing planet.

The operation and regulation of marine fisheries is discussed in **Chapter 8**. The chapter provides an overview of marine fisheries management, focussing on divergent approaches (single-species, multi-species, ecosystem-based) that have been applied – sometimes successfully and sometimes not. A history of international fisheries law and policy is then provided along with a summary of the organizations that oversee fisheries management. The primary international agreements that govern fishing are then discussed.

The **ninth chapter** explores the operation and regulation of international marine transportation. First, it begins with an overview of contemporary shipping and introduces marine transportation law and policy. Organizations that regulate marine transportation are then discussed along with the key international conventions and agreements dealing with the ship, shipping company and seafarer. The chapter then addresses international laws and policies related to maritime security (including piracy) and the environmental effects of shipping.

The Polar Regions (Arctic and Antarctic) warrant a separate treatment in **Chapter 10** since they have unique governance arrangements due to their remoteness, inhospitable climate and unique biological systems. While the Polar Regions are similar in climate, they are distinctive in almost all other respects, be they biological structure, governance or human use. Chapter 10 introduces the Antarctic and Southern Ocean and the Antarctic Treaty System that governs these regions. Governance of the Arctic Regions is then discussed along with the key institutional arrangements that assist with Arctic management.

Chapter 11 introduces the international law and policy instruments that apply to ocean energy and mining activities. It first differentiates energy and mining opportunities between the continental shelves and the deep seabed; it then discusses the role of the LOSC in energy and mineral development. Organizations that oversee energy and mining activities are then reviewed. A brief summary of marine energy and mineral resources and their management in the continental shelf and deep-sea areas ensues.

Integrated approaches to ocean management are elicited in **Chapter 12.** These approaches are loosely defined as tools to assist marine decision-making and the analysis of policies. Coastal management, ecosystem approaches to management, marine protected areas, systematic conservation planning, marine spatial planning and cumulative effects are analysed.

Terminology used in this book

Lawyers, policy analysts, environmental planners, engineers, diplomats and bureaucrats have their own lexicons related to ocean governance. This text borrows from all of these disciplines and uses the following terms based on recent practice:

States are equivalent to countries, nations, nation-states and economies, including states that are still in the process of achieving full recognition by the international community.

Marine and **ocean** are used interchangeably. However, 'maritime' has specific connotations with human use, particularly the marine transportation sector and is used only in this context.

Law as an umbrella term may either connote domestic or international law and includes treaties, legislation, rules and regulations.

Marine/ocean policy is the stated intent and subsequent actions, usually by governments, to influence ocean-related development, activities and outcomes.

Governance is the design, creation and implementation of laws or policies.

Convention is synonymous with an international agreement or treaty. When a particular convention is discussed, it is capitalized.

Law of the Sea Convention (LOSC) is used to abbreviate the United Nations Convention on the Law of the Sea as per common practice in the international law literature. The term 'UNCLOS', often used in marine policy literature, is synonymous here with 'LOSC'.

Further reading

Glavovic, B. (2008) 'Ocean and coastal governance for sustainability: Imperatives for integrating ecology and economics', in M. Patterson and B. Glavovic (eds.), *Ecological Economics of the Oceans and Coasts*, Edward Elgar, Cheltenham, UK.

Russell, B. (1950) *Unpopular Essays*, Routledge, New York.

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Acronyms and abbreviations

AAC	Arctic Athabaskan Council
ABNJ	areas beyond national jurisdiction
ACAP	Agreement on the Conservation of Albatrosses and Petrels
ACAP	Arctic Contaminants Action Program
ACIA	Arctic Climate Impact Assessment
AEPS	Arctic Environmental Protection Strategy
AHP	analytic hierarchy process
AIA	Aleut International Association
AMAP	Arctic Monitoring and Assessment Programme
AOP	Australia's Ocean Policy
APFIC	Asia-Pacific Fisheries Commission
ARREST	International Convention on Arrest of Ships
ASBAO	Regional Convention on Fisheries Cooperation among African States Bordering the Atlantic Ocean
ASOC	Antarctic and Southern Ocean Coalition
AT	Antarctic Treaty
ATCM	Antarctic Treaty Consultative Meeting
ATS	Antarctic Treaty System
BAS	British Antarctic Survey
BP	British Petroleum
BUNKERS	International Convention on Civil Liability for Bunker Oil Pollution Damage
BWM	International Convention for the Control and Management of Ships' Ballast Water and Sediments
CAFF	Conservation of Arctic Flora and Fauna
CAM	coastal area management
CBD	Convention on Biological Diversity
CCAMLR	Convention on the Conservation of Antarctic Marine Living Resources
CCAS	Convention for the Conservation of Antarctic Seals
CCSBT	Commission for the Conservation of Southern Bluefin Tuna
CDM	clean development mechanism
CDQ	community development quotas
CE	cumulative effects
CECAF	Committee for the Eastern Central Atlantic Fisheries
CEP	Caribbean Environment Program
CEP	Committee for Environmental Protection

CER	corporate environmental responsibility
cif	cost, insurance and freight
CITES	Convention on International Trade in Endangered Species of Wild Fauna and Flora
CLC	International Convention on Civil Liability for Oil Pollution Damage
CLCS	Commission on the Limits of the Continental Shelf
CLEE	Convention on Civil Liability for Oil Pollution Damage Resulting from Exploration for and Exploitation of Seabed Mineral Resources
CLL	International Convention on Load Lines
CMI	Comité Maritime International
CMS	Convention on the Protection of Migratory Species of Wild Animals
COFI	Committee on Fisheries
COLREG	Convention on the International Regulations for Preventing Collisions at Sea
COLTO	Coalition of Legal Toothfish Operators
COMNAP	Council of Managers of National Antarctic Programs
COP	conference of the parties
CPPS	Permanent Commission on the South Pacific
CPs	consultative parties
CPUE	catch per unit effort
CRM	coastal resource management
CSI	Container Security Initiative
CZM	coastal zone management
DSM	deep-seabed mining
DFO	Fisheries and Oceans Canada
DMC	dangerous maritime cargoes
dwt	dead weight tonnes
EA	enterprise allocations
EAF	ecosystem approach to fisheries
EAFM	ecosystem approach to fisheries management
EAGGF	European Agricultural Guidance and Guarantee Fund
EAM	ecosystem approaches to management
EAP	Action Plan for the Protection, Management and Development of the Marine and Coastal Environment of the Eastern African Region
EbA	ecosystem-based adaptation
EBSA	ecologically and biologically significant area
EEC	European Economic Community
EEDI	Energy Efficiency Design Index
EEZ	exclusive economic zone
EFR	ecological fiscal reform
ELECRE	elimination and choice expressing reality
EPPR	Emergency Prevention, Preparedness and Response
ERDF	European Regional Development Funds
ERFEN	Protocol on the Regional Program for the Study of the El Niño phenomenon in the Southeast Pacific
ESF	European Social Fund
ESG	environmental, social and governance
ETR	ecological tax reform
EU	European Union

EUR	Euro
EVI	economic vulnerability index
FAL	Convention on Facilitation of International Maritime Traffic
FAO	Food and Agriculture Organization
FCWC	Fishery Committee of the West Central Gulf of Guinea
FFA	South Pacific Forum Fisheries Agency
fob	free on board
FSC	flag state control
GBRMPA	Great Barrier Reef Marine Park Authority
GDP	gross domestic product
GDS	geographically disadvantaged states
GFCM	General Fisheries Council for the Mediterranean
GGI	Gwich'in Council International
GHG	greenhouse gasses
GIS	geographic information system
GPA	Global Programme of Action
GPS	global positioning system
GSPs	green shipping practices
GT/G.T./gt	gross tonnage
GW	gigawatts
HAFS	International Convention on the Control of Harmful Anti-Fouling Systems on Ships
HAI	human assets index
IAATO	International Association of Antarctica Tour Operators
I-ATTC	Inter-American Tropical Tuna Commission
IBSFC	International Baltic Sea Fisheries Commission
ICC	Inuit Circumpolar Council
ICCAT	International Commission for the Conservation of Atlantic Tunas
ICES	International Council for the Exploration of the Sea
ICJ	International Court of Justice
ICM	integrated coastal management
ICNAF	International Commission for the Northwest Atlantic Fisheries
ICRW	International Convention for the Regulation of Whaling
ICZM	integrated coastal zone management
IFQ	individual fishing quotas
ILA	International Law Association
ILC	International Law Commission
ILO	International Labour Organization
IM	integrated management
IMO	International Maritime Organization
IOPC	Funds International Oil Pollution Compensation Fund
IOS	Indian Ocean Sanctuary
IOTC	Indian Ocean Tuna Commission
IPHC	International Pacific Halibut Commission
IPOA	international plans of action
ISA	International Seabed Authority
ISMC	International Safety Management Code
ISO	International Organization for Standardization
ISPS	International Ship and Port Facility Security Code

ITLOS	International Tribunal for the Law of the Sea
IUCN	International Union for the Conservation of Nature
IUU	illegal, unreported, and unregulated
IVQ	individual vessel quotas
IWC	International Whaling Commission
kg	kilogram
lb	pound
LDC	Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter
LDS	landlocked developing states
LLDCs	landlocked developing countries
LLMC	Convention on Limitation of Liability for Maritime Claims
LOSC	Law of the Sea Convention
LSMPA	large scale marine protected area
M/V	or MV motor vessel
MAP	Mediterranean Action Plan
MARPOL	International Convention for the Prevention of Pollution from Ships
MAUT	multi-attribute utility theory
MBI	market-based instruments
MBTA	Migratory Bird Treaty Act
MCDA	multi-criteria decision analysis
MEPC	Marine Environment Protection Committee
MLC	Maritime Labour Convention
MPA	marine protected area
MSC	Marine Stewardship Council
MSC	Maritime Safety Committee
MSP	marine spatial planning
MSY	maximum sustainable yield
MW	megawatts
NAFO	North Atlantic Fisheries Organization
NAMMCO	North Atlantic Marine Mammal Commission
NASCO	North Atlantic Salmon Conservation Organization
NEAFC	North East Atlantic Fisheries Commission
nmi	nautical mile
NOAA	National Oceanic and Atmospheric Administration
NDC	nationally determined contributions
NPA	National Programmes of Action
NPAFC	North Pacific Anadromous Fish Commission
NT/N.T./nt	net tonnage
OCA/PAC	Oceans and Coastal Areas Programme Activity Centre
OLDEPESCA	Latin American Organization for the Development of Fisheries
OPA	Oil Pollution Act
OPC	Oil Pollution Convention
OPRC	International Convention on Oil Pollution Preparedness, Response, and Co-operation
OSPAR	Convention for the Protection of the Marine Environment of the Northeast Atlantic
OSY	optimum sustained yield
P&I	Clubs protection and indemnity clubs

PAME	Protection of the Arctic Marine Environment
PCA	Permanent Court of Arbitration
PCBs	poly-chlorinated biphenols
PERSGA	Programme for the Environment of the Red Sea and Gulf of Aden
PICES	North Pacific Marine Science Organization
PM	particular matter
PROMETHEE	preference ranking organizational method for enrichment evaluation
PSC	Pacific Salmon Commission
PSC	port state control
PSI	Proliferation Security Initiative
PSSA	particularly sensitive sea area
PSU	practical salinity units
RAIPON	Russian Arctic Indigenous Peoples of the North
ReCAAP	Regional Cooperation Agreement on Combating Piracy and Armed Robbery against Ships in Asia
RFMOs	regional fisheries management organizations
RMP	Revised Management Procedure
RMS	Revised Management Scheme
ROPME	Regional Organization for the Protection of the Marine Environment
ro-ro	roll on/roll off
SAP	BIO Strategic Action Plan for the Conservation of Marine and Coastal Biodiversity in the Mediterranean
SAR	Convention International Convention on Maritime Search and Rescue
SC	Saami Council
SCAR	Scientific Committee on Antarctic Research
SCP	systematic Conservation Planning
SDC	Seabed Disputes Chamber
SDR	Special Drawing Rights
SDWG	Sustainable Development Working Group
SEAFO	South East Atlantic Fisheries Organization
SECAs	sulphur emission control areas
SEEMP	Ship Energy Efficiency Management Plan
SIDS	small island developing states
SIOFA	South Indian Ocean Fisheries Agreement
SOLAS	International Convention for the Safety of Life at Sea
SOS	Southern Ocean Sanctuary
SPA	Protocol Concerning Specially Protected Areas and Biological Diversity in the Mediterranean
SPLOS	State Parties to the LOS Convention
SPR	spawning potential ratio
SPREP	South Pacific Regional Environmental Programme
SPRFMO	South Pacific Regional Fisheries Management Organization
SRCF	Sub-Regional Commission on Fisheries
SSB	spawning stock biomass
SSBR	spawning stock biomass per recruit
STCW	International Convention on Standards of Training, Certification and Watchkeeping for Seafarers
TAC	total allowable catch
TBT	tributyltin

TCF	trillion standard cubic feet
TEU	twenty-foot equivalent units
TURFs	territorial user rights in fisheries
UBC	University of British Columbia
UK	United Kingdom
ULCC	ultra large crude carrier
UN	United Nations
UNCED	United Nations Conference on Environment and Development
UNCHE	United Nations Conference on the Human Environment
UNCITRAL	United Nations Commission on International Trade Law
UNCLOS	I 1958 United Nations Conference on the Law of the Sea
UNCLOS	II 1960 United Nations Conference on the Law of the Sea
UNCLOS	III 1982 United Nations Conference on the Law of the Sea
UNCLOS	United Nations Convention on the Law of the Sea
UNCTAD	United Nations Conference on Trade and Development
UNEP	United Nations Environment Program
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNFCC	United Nations Framework Convention on Climate Change
UNFSA	UN Fish Stocks Agreement
UNGA	United Nations General Assembly
UNGAR	United Nations General Assembly Resolution
UNHCR	United Nations High Commission for Refugees
UNSCR	United Nations Security Council
USA	United States of America
USSR	Union of Soviet Socialist Republics
UV	ultra-violet
VLCC	very large crude carrier
VOCs	volatile organic compounds
WACAF	Abidjan Convention for Co-operation in the Protection and Development of the Marine and Coastal Environment of the West and Central African Region
WCED	World Commission on Environment and Development
WCPFC	Western and Central Pacific Fisheries Commission
WECAFC	Western Central Atlantic Fisheries Commission
WiP	with policy
WMD	weapons of mass destruction
WRC	Nairobi International Convention on the Removal of Wrecks
WSSD	World Summit on Sustainable Development
WTO	World Trade Organization
WWF	World Wildlife Fund

Chapter 1

An introduction to the world ocean

Human impacts and trends

Key topics

- The ocean is biogeochemically downstream from terrestrial environments, as such, most human activities eventually impact the world ocean.
- Threats to biodiversity can be broadly categorized as a result of overharvesting, pollution, habitat loss, introduced species, ocean acidification and global climate change. Threats can cumulatively interact with negative consequences.
- Until recently, different human impacts on the world ocean were managed in isolation, resulting in seemingly rational management actions that sometimes have unintended consequences.

Introduction

Humanity requires the services of the world ocean in a multitude of ways. For centuries, the oceans were envisaged as immutable and immune to human activities (see Box 1.1 and Box 1.2), and, until the 20th Century, the regulation of human activities affecting the world ocean was widely perceived as unnecessary and unachievable, given the absence of state ownership or hard boundaries in the marine environment. In addition, unlike the human footprint on the terrestrial realm, which is often easy to observe, most marine environments are hidden from us and therefore out of sight and out of mind (see Box 1.3).

The scale of the human footprint on the world ocean is enormous. Many stocks of once globally abundant fishes such as cod, herring and tuna have since become 'commercially extinct;' i.e. no longer commercially fishable due to their greatly reduced numbers. Over one million whales were harvested during the 19th and 20th Centuries; to date, only the eastern Pacific grey whale has recovered to near pre-exploitation levels. The global seabird population has declined by 69.7 per cent since 1950; over the past 40 years, the abundance of marine vertebrates (fish, seabirds, sea turtles and marine mammals) has declined by an average of 22 per cent (Paleczny et al., 2015; Sumalia et al., 2016). Elevated levels of pollutants are found in most marine species, even those living in the Polar Regions. Ocean temperatures are rising – aggravated by the addition of greenhouse gasses from the combustion of fossil fuels – with many deleterious effects. Tens of thousands of square kilometres of coral reefs have bleached (lost their photosynthetic symbionts), with large associated die-offs in recent years. Important breeding, feeding, mating and resting areas for migratory species have been affected by human activities. The continuing human-induced degradation of marine ecosystems is as wide and deep as the ocean itself (Roberts, 2013).

Box 1.1 Importance of the oceans

Globally, the oceans are the:

- Main reservoir of water: 71 per cent of the earth's surface is covered by oceans; less than 0.5 per cent is freshwater.
- Main place for organisms to live; they comprise over 99 per cent of the inhabitable volume of the 'earth'.
- Main planetary reservoir of O_2.
- Possible main planetary producer of O_2 from phytoplankton.
- Planetary thermal reservoir and regulator.
- Medium for longitudinal heat transfer and circulation.
- Major reservoir of CO_2 especially in HCO_3-, CO_3 = forms.
- Habitat for enormous diversity of living organisms, from bacteria to whales.
- Reservoirs of enormous resource potential, both renewable and non-renewable, oil, minerals, etc.; also, the location of half the planet's carbon fixation.

Source: After Roff and Zacharias (2011)

Box 1.2 Marine goods and services that support humanity

Production services

- Food provision: Marine organisms for human consumption.
- Raw materials: Marine organisms for non-consumptive purposes, such as pharmaceuticals and genetic materials; non-living resources including minerals, sand and petroleum.

Cultural

- Identity/cultural heritage: Value associated with the marine environment, e.g. for cultural and spiritual traditions.
- Psychological health, leisure and recreation: Revitalizing the human body and mind through leisure activities ranging from visiting the beach to SCUBA diving with marine organisms in their natural environment.

Value

- Research value: Studying the many aspects of the ocean to gain a greater understanding of the world around us.
- Future unknown and speculative benefits: Currently unknown potential future uses of the marine environment and associated biodiversity.

Regulation services

- Gas and climate regulation: Balance and maintenance of the chemical composition of the atmosphere and oceans by marine living organisms and chemical processes.

- Flood and storm protection: Dampening of environmental disturbances by biogenic structures such as coral reefs, kelp beds and mangroves.
- Bioremediation of waste: Removal of pollutants through storage, dilution, transformation and burial.

Supporting services

- Nutrient cycling, including carbon sequestration: Storage, cycling and maintenance of nutrients by living marine organisms and physical processes.
- Biologically mediated habitat: Habitat that is provided by marine organisms.
- Resilience/resistance: Extent to which ecosystems can absorb recurrent natural and human perturbations and continue to offer the above-listed services without degrading or unexpectedly switching to alternate states.

Source: Beaumont et al. (2007) and Ruiz-Frau et al. (2011)

Box 1.3 Human perceptions on ocean health

A recent review of research into the public attitudes and perceptions on threats to the world ocean found that regardless of which state the survey was conducted in, there was agreement (70 per cent) that oceans are threatened by human activities, and 45 per cent of respondents ranked these threats as high or very high. The review collated over 32,000 respondents across 21 nations and found that while the public ranked the threats of pollution and overfishing respectively as the greatest threats, there were regional disparities based on past events (e.g. oil spills in Scotland), media campaigns (e.g. ocean plastics) and differences in opinion between the public and marine scientists (e.g. the threats of climate change). In addition, most respondents wanted to see more of the world ocean protected but were unsure how much is already protected or what the socio-economic costs of additional protection would be.

Public opinion surveys are important to policy-makers as they identify electorate priorities, where to focus education and awareness initiatives and how management and conservation programs may be improved with limited available resources.

Source: Lotze et al. (2018)

Although human capacity for massive disturbance in marine environments has been known at least since the extinction of the Steller's Sea Cow in 1868, the plight of the oceans did not become a public concern until the appeals in the 1950s and 1960s by authors such as Rachel Carson (1962) and Jacques Cousteau. Since the 1970s, the steadily increasing output of books, films and television series along with the establishment of organizations such as Greenpeace have resulted in heightened public awareness and concern. Inter-governmental efforts to better manage the marine environment began in earnest with international conventions such as the London Dumping Convention (1972), the International Convention for the Prevention of Pollution from Ships (1978), the United Nations Convention on the Law of the Sea (1982), and the Convention on Biological Diversity (1992). More recently, under the Paris Agreement (2015) of the United Nations Framework Convention on Climate Change (UNFCCC, 1992), states have gradually begun to recognize the importance of the global ocean in meeting targets and 'nationally determined contributions' (NDCs).

Notwithstanding these efforts, human uses and impacts on the ocean continue unabated and the diversity of life in our oceans is now being dramatically altered by rapidly increasing and potentially irreversible human activities. Approximately 40 per cent of the world's population and 60 per cent of the world's economic production are concentrated in a 100km swath along the world's coasts. Twenty-one of the world's 33 mega cities are coastal, and it is estimated that by 2020 up to 75 per cent of the world's 7.5 billion people may be living within 60km of the coastal zone (Martínez et al., 2007; UN, 2017).

Human impacts on the oceans can be broadly categorized as a result of overharvesting, pollution, habitat loss, introduced species and global climate change/ocean acidification. The following sections provide a brief discussion of these impacts on the marine environment. It should be noted, however, that these impacts do interact, often in ways detrimental to ocean health. These negative interactions are termed cumulative or synergistic effects. The seemingly rational regulation and management of individual sectors generally fails to consider the collective impacts of other sectoral activities on either the environment, public health or human use of the resource, sometimes resulting in unintended negative consequences to ecological systems and human well-being. Cumulative effects may be localized (e.g. pollution discharge) or global (e.g. climate change) and may be acute (e.g. overharvesting a fishery) or enduring (e.g. disposed radionuclides).

Overharvesting

Unsustainable harvesting is perhaps the most serious threat to marine environments worldwide. Overharvesting is not a new phenomenon in the oceans. Historically, human cultures either removed particular species and moved on to harvesting others, or other areas, or had to develop methods of regulating the timing and amount of harvest in order to avoid over-exploitation. With the advent of the industrial revolution in the mid to late 1800s came the increasing power to fish in ways hitherto unheard of: Steam-trawling the seafloor, fishing further offshore, and mechanically winching on board large nets and animals. Species such as whales and offshore pelagic fish (tuna, swordfish) became accessible to those willing to take the risks. Most whale species have been reduced to levels where they are considered endangered or threatened. Most populations of palatable fish stocks, particularly the larger fish, have been seriously depleted (e.g. Pauly et al., 1998; Worm et al., 2006).

From a western perspective, the importance of marine fisheries to global food security is often overlooked. In North America and many parts of Europe, seafood is simply another source of protein for consumers to choose. Elsewhere, over one billion people depend on seafood as their primary source of protein, and 3.1 billion people rely on seafood for at least 20 per cent of their protein. Fishing has recently been estimated to employ 34 million people in full- or part- time jobs, producing 81.5 million tonnes of seafood in 2014. The first-sale value of the world's fisheries is estimated at $US 100 billion (FAO, 2017).

Be that as it may, the sustainability of marine capture fisheries has recently been shown to be at risk. The Food and Agriculture Organization of the United Nations (FAO) currently reports that 31.4 per cent of the world's fisheries are now overexploited, depleted or recovering, 58.1 per cent are fully exploited and only 10.5 per cent under exploited or moderately exploited (FAO, 2017). For context, in 1974, the corresponding percentages were 10, 50 and 40 per cent, respectively. Global catches peaked in 1996 and, for the past

15 years, have been relatively stable at a level approximately 10 per cent less than 1996 (FAO, 2017). The European Commission has determined that for the European fish stocks where stock status is known, only a third are currently managed for sustained future production (CEC, 2008).

The status of high seas stocks (i.e. those beyond national jurisdictions) is even less well known. Of the 17 Atlantic fish stocks managed under the North Atlantic Fisheries Organization (NAFO), six are collapsed and only two are considered to be sustainable (NAFO, 2016). Atlantic cod biomass is estimated at 6 per cent of historical levels and North Sea cod stocks are depressed to such a degree that recently fishers have been unable to harvest enough to meet their allowable catches, and each year further reductions are recommended. In 2018, the International Council for the Exploration of the Sea (ICES) recommended a 47 per cent reduction to catches of cod in the North Sea, eastern English Channel and Skagerrak Region (ICES, 2018).

A closer look at certain aspects of marine capture fisheries shows that 50–70 per cent of pelagic predators (i.e. tunas, swordfish) have been removed by fishing. Fishing pressure continues to shift towards lower trophic levels as apex predators decline; this has been termed 'fishing down marine food webs' (Pauly et al., 1998). Furthermore, the global fishing fleet is far larger than what necessity dictates, especially since technological innovations increase the ability to catch fish. This surplus fishing capacity is underwritten by $US 30–40 billion in annual subsidies by most fishing nations, thus providing no incentive to reduce fishing effort (Sumaila et al., 2016). In addition, evidence suggests that fishing efforts from port-based fisheries have increased over the past three decades at upwards of 3 per cent per decade. As such, more fishers are chasing fewer fish; if one were to imagine the global fishing fleet as a country, it would be the 18th largest oil-consuming nation on earth.

Another serious problem is the unintended harvest of species. An estimated 20 per cent of the global fisheries' catch constitutes unwanted by-catch that is discarded. Shrimp trawl fisheries produce the largest by-catch and small pelagic fisheries the least. By-catch also consists of the incidental take of endangered marine mammals, turtles and seabirds although recent advances in gear technology are reducing these impacts.

There are signs that some fisheries are on a path to sustainability. In the United States and New Zealand, the numbers of overfished stocks have been steadily improving, from over 20 per cent a decade ago to about 15 per cent currently (Fisheries New Zealand, 2017; NOAA, 2017). Canadian northern cod stocks off of Newfoundland and Labrador – fished to collapse between the 1960s and 1990s – have seen recent growth rates of up to 30 per cent, indicating that even stocks thought to be fished past the point of no return can recover if protected (Rose and Rowe, 2015).

Pollution

Pollution – from a myriad of sources – has impacted every marine system on earth. As a result of global transport mechanisms, contamination has spread to all continents and depths. Indigenous human populations in Arctic areas have the highest contaminant levels of any people on earth, a result of ingesting marine fish and mammals which bioaccumulate toxins due to their high trophic levels. Up to 90 per cent of pollution entering the ocean is derived from land-based sources; the remainder is a result of deliberate or accidental dumping. It has been assumed that land-based sources of pollutants reaching the oceans is primarily via rivers; however, transport of pollutants through the atmosphere to the oceans is also significant, particularly for metals such as mercury. Sources of pollution in the ocean may be point source (from a single source, such as an industrial facility or sewage discharge) or non-point source (where the exact source is unknown, e.g. agricultural runoff).

Most researchers recognize the primary types of pollution as biological (pathogens), toxic substances (contaminants), nutrients, marine debris, hydrocarbon spills and underwater noise. Certain scientists also suggest that ocean acidification, as a result of climate change, is a type of pollution.

Biological pollution (pathogens) includes increased levels of bacteria, viruses and parasites, occurring naturally or introduced as a result of human activities. They may reside in seawater, the seabed or live on or within marine organisms. Anthropogenic (human) sources of pathogens in the ocean include sewage and agricultural runoff. Pathogens affect human health through direct ingestion (e.g. shellfish, seawater when swimming), skin contact or inhalation. Shellfish are known to transmit gastroenteritis and hepatitis, as well as cholera and typhoid fever. Marine mammals are known to have been infected by terrestrial animal diseases, likely transmitted from freshwater into the ocean. Fish hatcheries and aquaculture operations may also inadvertently introduce pathogens into the environment. Climate change may exacerbate the effects of certain marine pathogens mainly through increased temperatures providing optimal growth conditions (Burge et al., 2014).

Over the past 100 years of industrial chemistry, hundreds of *synthetic compounds* have found their way into the world ocean. Thousands of new compounds are created each year with sometimes unknown implications should the compound be released in the environment and without a regulatory regime to apply precaution to their use. Synthetic compounds have been categorized in many ways, but generally they can be separated into persistent organic pollutants (POPs), pesticides, pharmaceuticals, flame retardants and other substances captured under the general heading of 'contaminants of emerging concern' (or CECs). Unlike pathogens, the consequence of synthetic compounds on the environment and human health are subtler, including long-term bio accumulative effects leading to potential chronic illness and cumulative/delayed effects.

Persistent organic pollutants (POPs) capture a large group of chemicals that are similar in that they are halogenated (containing fluorine [F], chlorine [C], bromine [Br], iodine [I] and astatine [At]) organic compounds that degrade very slowly in the environment, bioaccumulate in fatty tissues and are highly toxic. Persistent organic pollutants have a wide range of effects including changes in behaviour, development, growth and reproductive capacity.

Uses of POPs have varied since their introduction, but common uses have included pesticides (e.g. DDT), industrial chemicals (e.g. PCBs), solvents and pharmaceuticals. Over time, the use of the more persistent POPs has slowly been phased out through international agreements or individual nations passing bans, resulting in a gradual decrease of these substances in the marine food chain. However, while the compounds replacing the 20+ banned POPs are certainly less persistent and toxic, they continue to impact marine food webs, often in new and unknown ways (Desforges et al., 2015).

'First generation' pesticides, such as DDT, were banned by many countries in the 1970s and internationally in 2001 due to persistence and toxicity. 'Second generation' pesticides consist primarily of organophosphates and carbamates, which are cholinesterase inhibitors, meaning they inactivate a key enzyme necessary for nervous function. These compounds break down much quicker in the environment; however, they can still be toxic to marine life. 'Third generation' pesticides either work via mimicking juvenile hormones so the insect cannot reach adulthood or through preventing insect larvae from moulting and are the safest pesticide for marine environments.

The effects of pharmaceuticals on the marine environment is a relatively new branch of study. Humans use over 6,000 different prescription and non-prescription drugs that enter the marine environment through either direct disposal (e.g. flushing unused medicines) or human wastes. Approximately 90 per cent of oral drugs pass through the body that in turn pass unaffected through sewage treatment systems and ultimately enter the oceans. While there have been demonstrated effects on marine life from certain pharmaceuticals, such as birth control pills reducing reproductive success in invertebrates due to endocrine disruption or development of antibiotic resistance in marine microbes, the overall impacts of drug inputs into the marine environment is poorly understood. However, early evidence suggests that many drugs are biologically active in marine environments at very low concentrations and have the capacity to affect behaviour and reproductive success of marine biota that reside near human settlements (Mezzelani et al., 2018). In addition to pharmaceuticals, personal care products such as cosmetics, antibacterial soaps, lotions and insect repellents are not removed in the sewage treatment process and exhibit many of the same properties as prescription drugs.

Lastly, CECs include a range of compounds that include: Flame retardants used in furniture, carpets, clothing and toys; plasticizers that increase the flexibility or fluidity of materials; solvents capable of dissolving other substances; surfactants that break down the interface between water and oils and therefore aid cleaning substances; and nanoparticles, which are substances smaller than one hundred nanometres used in a range of industrial processes and consumer products.

Metals are naturally occurring elements that, when increased above background levels by human activities, may have significant impacts on environmental and human health. Certain metals such as copper (Cu) and zinc (Zn) are necessary to sustain life; however, they can be toxic at higher concentrations. Other metals, including cadmium (Cd), chromium (Cr), lead (Pb) and mercury (Hg) have no role in biological processes but are highly toxic. Metals are introduced into the ocean via land-based processes (mining, industrial processes), the atmosphere (e.g. coal combustion and cement production releasing mercury) and anti-fouling compounds used to keep vessels and other marine structures free of marine plants and invertebrates.

Metals affect the marine environment and humans in a number of ways. First, metals – like certain other pollutants – bioaccumulate up the food chain and concentrate in top predators (e.g. tuna, swordfish, marine mammals, seabirds). Metals may bind to small clay and silt particles that are ingested by benthic (bottom) animals that are in turn consumed by other animals or stored in nearshore plant and marine algae tissues and in turn consumed. At certain concentrations, metals may affect respiration, digestion, nervous and reproductive system functions, leading to changes in behaviour, slower growth and reproductive failure (Burger et al., 2014).

While copper, cadmium, chromium and lead are generally found in localized areas (e.g. marinas, industrial areas) with localized impacts on environmental and human health, mercury is a global problem. Mercury is a potent neurotoxin that easily bioaccumulates, which is why most countries now recommend limits on tuna and swordfish consumption for children. In 2017, the global Minamata Convention on Mercury came into force given the rapid rise in mercury in marine biota and corresponding human health issues.

Nutrients include human inputs of nitrogen (N) and phosphorus (P) that exceed natural background levels resulting in algal blooms. Aside from upsetting the natural state of the food web, these algal blooms may also be toxic ('harmful algal blooms' or HABs) to marine animals (fish and shellfish) or humans or consume available dissolved oxygen creating 'dead zones' (hypoxia or anoxia) due to a lack of oxygen for other species, a process termed 'eutrophication'. Harmful algal blooms have expanded in recent decades, primarily due to increasing coastal human populations and the resultant creation of N from agriculture (animal wastes and fertilizers), sewage and the deposit of airborne N from fossil fuel combustion into rivers and oceans. Climate change may also be exacerbating HABs due to increases in water temperatures, fostering the increased growth of primary producers such as phytoplankton.

Marine debris is any intentionally or unintentionally discarded solid manufactured item in the marine environment. NOAA estimates that approximately 600 million kilograms of trash enters the ocean each year, 60–80 per cent of which are plastics (NOAA, 2016). Polystyrene foam or 'styrofoam' is another major source of marine debris that tends to float while the majority of plastics sink. The majority of marine debris (about 70 percent) is found on the seabed while the remainder is split evenly between floating on the ocean surface or found on beaches.

The primary types of marine debris are domestic materials (e.g. shopping bags, plastic bottles), industrial materials and fishing gear. Plastics are durable. Though many will break apart into ever smaller pieces, becoming largely invisible 'micro-plastics', they will not break down chemically for hundreds of years. Additionally, plastics absorb and concentrate hydrophobic organic pollutants, thus acting as reservoirs and transport mechanisms for these substances. It is estimated the economic impact of ocean plastics is $US 13 billion dollars annually (UNEP, 2014). Approximately 80 per cent of marine debris comes from land-based sources.

Marine debris impacts the marine environment in a number of ways. Ingestion and entanglement are the primary effects on marine mammals, seabirds, sea turtles and fishes. In highly populated coastal areas, plastics may affect plants and animals living on or in the seabed through accumulation resulting in degradation of habitat through smothering and toxicity. It is estimated that marine debris is responsible for up to 100,000 marine mammal deaths annually and 90 per cent of seabirds have plastics in their bodies. Ingested plastics inhibit growth and reproduction through lacerating or displacing food in animals' stomachs or transferring toxic substances to the host. Lost or abandoned fishing gear can continue to 'ghost fish' where fish persist to be caught in nets that may have been adrift for many years (Gall and Thompson, 2015).

Hydrocarbon (oil) spills are a unique type of pollution. Hydrocarbon spills occur as a result of the transport of different forms of oil or the exploration and development of offshore oil resources. Approximately half of the world's oil production (approximately 1.6 billion tonnes) is transported by sea. Much of this transport is over long distances in large tankers that pass through straits and transit along coastlines. This means of transport has resulted in many serious accidents and spills over the past several decades; however, due to improvements in safety, currently about 12 per cent of oil entering the ocean is a result of transportation accidents with the remainder from primarily land-based sources. While the average number of major oil spills per year has dropped from 25 in the 1970s to 3 today, smaller spills continue to impact fisheries, tourism and coastal economic activities.

Marine environments also currently produce one third of the world's hydrocarbons and have the potential for significant expansion. The shallow marine (< 400 metres) and deep marine (> 400 metres) environments are estimated to contain 36 per cent and 11 per cent of the global distribution of oil-bearing reservoir rocks respectively, though much of the deep-sea has not yet been explored. Offshore reserves currently account for about 30 per cent of oil production globally. It is estimated that 300 billion barrels may yet be discovered in offshore areas.

In addition to shipping, a significant amount of oil is transported short distances by seabed pipelines from production wells to offshore or onshore facilities.

Habitat loss and degradation

Approximately 40 per cent of the world's population and 60 per cent of the world's economic production are concentrated in a 100km swath along the world's coasts. The confluence of such a significant proportion of the world's population on the narrow fringe separating the marine and terrestrial realms has significant ecological and socio-economic implications.

Habitat loss due to the destruction of larger vascular vegetation upon which many species depend for food and shelter is probably the most serious threat to biodiversity in terrestrial environments. Loss of marine habitat is primarily a concern in coastal nearshore and intertidal marine environments. Increasing pressure on coastal systems results from a combination of shipping and its attendant infrastructure, modification of natural coastlines, bottom contact fishing (e.g. trawling), recreational activities and increased land runoff – including nutrients and suspended solids. The types of habitats in these areas that can be 'lost' include marine macrophytes (kelp), mangroves, seagrasses, corals and other biotic communities (e.g. sponges, seapens, sea fans, aphotic corals). Abiotic habitats can be impacted as well, such as intertidal and estuarine mud flats and other areas that are dredged or subject to dumping. Habitat loss in deeper marine environments and the pelagic ocean is more difficult to determine as these habitats are primarily composed of either oceanographic (e.g. currents, gyres, fronts) or physiographic (e.g. seabed composition) structures and processes that are either not immediately impacted or more resistant to human activities (however, deep-seabed mining, should it progress to commercial operations, could radically alter deep-sea habitats). Loss of marine habitat is significant not only from an ecological perspective but increasingly from a socio-economic perspective as well. The interaction of human effects and natural marine processes is most evident in coastal waters where strategies to prevent habitat loss or restore habitats are integrated within coastal zone management initiatives.

Habitat loss can be rapid and its consequences significant. An estimated kilometre of Europe's coastline was developed each day between 1960 and 1995, resulting in a 50 per cent loss of coastal wetlands and seagrasses (Gibson et al., 2007). Much of the world's continental shelves have experienced habitat degradation due to bottom contact fishing (bottom trawling) where nets are dragged over the ocean floor, causing damage to sessile organisms such as corals and seapens, as well as flattening benthic (bottom) biogenic structures (e.g. oyster reefs). Furthermore, bottom contact fishing can result in the fine sands, silts and clays being resuspended in the water column for extended periods of time.

As natural resources become depleted or more difficult to access on land, the pressure to extract material from the oceans increases. The nearshore marine environment has long been utilized as a source for materials. For centuries, beaches and nearshore areas have been mined for gold (Au), tin (Sn), diamonds, sand and gravel. As mining and shipping technologies improve, exploration and development has radiated seaward onto the continental shelves. Offshore oil drilling was pioneered in the 1930s in the Gulf of Mexico and the first offshore windfarm established in Denmark in the 1990s. Currently, almost one third of the world's petroleum is produced from continental shelves. This output is likely to continue to increase as sea ice retreats in the Polar Regions, opening up new areas for exploration and development.

It is estimated that deep-sea areas (outside of continental shelves) contain two thirds of the world's accessible mineral wealth. For the past 60 years, anticipated mineral shortages combined with increased commodity prices have created periods of interest in mining the mineral-rich deep-sea. However, these prognosticated shortages have failed to transpire, due to a combination of increased metal recycling, refinements in industrial processes and new terrestrial discoveries. Additionally, the technological challenges of mining the deep seabed have proven more difficult than some anticipated. To date, lower than expected commodity prices combined with higher than expected technological costs have made it cost prohibitive for deep-sea mining to become economically attractive to conventional mining firms, though interest remains with some states and newcomer companies. Long-term demand – based on forecasted development of land-based deposits – suggests that marine deposits high in cobalt (for batteries) could become commercially viable at some point in the future.

A final potential future impact on continental shelf environments is methane hydrates. Produced by the decomposition of organic materials, methane hydrates consist of molecules of natural gas (predominately methane) trapped in ocean sediments by a combination of low temperature and high pressure. They are generally found in water depths greater than 400–500 metres and up to 1,100 metres below the seafloor. One cubic metre of hydrate is equivalent to 164 cubic metres of methane at atmospheric pressure and temperature and the energy potential of the world's reserves is estimated between 10^5 and 10^8 trillion standard cubic feet (TCF). These astounding numbers make the energy potential of methane hydrates greater than all the world's known coal, oil and natural gas reserves.

Introduced species

Species introductions (also termed invasive, exotic and non-native species) likely have occurred since humans first used the oceans for exploration and trade. There is evidence that many species believed to be native were introduced through marine transportation prior to the industrialized era. More recently, species introductions are facilitated through trade, travel and transport because of globalization and population growth. The rate of species introductions has grown by an order of magnitude over the past three decades. Currently, 1,369 non-native species are reported in European seas, whereas a decade earlier there were only 737. California, with a much smaller coastline, is host to 235 known invasive species (Williams et al., 2013).

The ballast water of ships appears to be the main mode of travel (vector) for introduced species, and impacts are generally observed mainly in coastal waters and estuaries. Once established, introduced species can have a number of detrimental impacts on their new environments including out-competing native species for food and habitat, introducing new pathogens, hybridizing with native species or disruption of entire food webs. Some introduced species, such as the Japanese oyster (*Crassostrea gigas*), have been intentionally introduced throughout the world for commercial aquaculture purposes and, although they displace native species, are generally regarded as a net benefit. Others, such as the American comb jellyfish (*Mnemiopsis leidyi*), have had disastrous consequences. The jellyfish arrived in the Black Sea in 1982 in the ballast water of ships and – without predators or competitors – quickly consumed most of the available zooplankton, leading to the collapse of fisheries and significant economic impacts. By the 1990s, the comb jellyfish accounted for 90 per cent of the biomass in the Black Sea.

Some of these invasive species can have dramatic local socio-economic effects. Different species of jellyfish have had substantial impacts on fisheries and even coastal human recreation. Outbreaks of jellyfish have now been reported worldwide due to species invasions, overharvesting (removing jellyfish predators), nutrient enrichment (increasing primary production resulting in increases in available food) and increases in local water temperature. Most species introductions are less obvious, found in the phytoplankton and zooplankton, with ecosystem impacts throughout the food web.

Global climate change

There is no doubt that the earth's climate has changed throughout history and that cyclical changes transpired long before humans dominated the planet. Global climate changes have been responsible for mass extinctions in the past and will likely result in future extinction events. During the Quaternary period, sea levels deviated as much as 85m, which inhibited the evolution of established marine communities in coastal and shelf environments. Unquestionably, humans have impacted global temperatures; since the 1980s, there has been considerable debate on differentiating the natural and anthropogenic contributions to climate change. Recent evidence, however, concludes that climate change since the 1900s is human-induced, and it is estimated that the oceans have lost 2 per cent of their oxygen since 1960. (Gilbert, 2017)

Human activities that influence change include the release of carbon dioxide through the burning of fossil fuels and large-scale deforestation, which decreases the removal rate of CO_2 from the atmosphere. Changes in ocean temperatures impact marine biological communities in various ways, including changes in geographical distribution, behaviour and life history (e.g. reproduction, growth and dispersal). It is estimated that 892 commercially important fish species are shifting into new territories at a rate of 70km per decade and that the world's exclusive economic zones (EEZs) are anticipated to receive up to five new fish stocks by the end of the century (Poloczanska et al., 2013). Coral reefs, for example, are particularly vulnerable to ocean warming, and even minor temperature increases may cause many coral species to expel their algal symbiotic zooxantheallae (termed 'bleaching') that provide energy through photosynthesis, ultimately causing coral death if temperatures stay elevated and the zooxantheallae do not return. There is also evidence that the ranges of some species have shifted towards the north and south poles as temperatures increase in equatorial and temperate waters (Pinsky et al., 2018).

The global shipping industry emits significant amounts of greenhouse gasses. The International Maritime Organization (IMO) estimates that shipping was responsible for 796 million tonnes of CO_2 emissions, or about 2.2 per cent of global emissions in 2012 (IMO, 2014). If the global shipping industry were a country, it would be the sixth largest emitter of CO_2 after the US, China, Russia, India and Japan. Shipping moves approximately 70 per cent of global freight while producing around 15 per cent of global CO_2 emissions. Without additional regulations on CO_2 emissions or the adoption of more efficient shipping technologies, shipping emissions are forecasted to increase by a factor of 2 to 3 by 2050. On the other hand, if technical and operational improvements are implemented, they could reduce CO_2 emissions by 25 per cent to 75 per cent below current levels.

Discussion questions

- Of the five main threats to the world ocean, which is the most serious to the continued ecological function of the ocean and the goods and services it provides to humanity?
- Of the five main threats to the world ocean, which do you believe individual states (countries) can address independently, and which will require states to work together to address?
- If status quo management of the world ocean continues, what might be the impacts to humanity in 50 years?

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